A bone remodelling model including calcium-phosphorus homeostasis

Javier Martínez Reina[1], José Luis Calvo Gallego[1], Rocío Ruiz Lozano[1], Peter Pivonka[2], Ralph Müller[3]

[1] Universidad de Sevilla, España, jmreina@us.es
[2] Queensland University of Technology, Australia
[3] ETH Zürich, Suiza

Introduction

Calcium-phosphorus (Ca/P) homeostasis and bone remodelling are closely linked, yet rarely integrated. We present a multiscale framework coupling endocrine regulation with cellular dynamics bone tissue mineralization to analyze how chronic kidney disease (CKD) and primary hyper-parathyroidism (pHPT) may impact bone metabolism leading to osteoporosis. This approach has enabled quantitative evaluation of dietary and pathological effects of both conditions.

Methods

The model presented here couples a mechanistic, cell-based bone remodelling model [1] with a systemic Ca/P homeostasis model [2]. The latter includes bidirectional interactions between endocrine regulation, particularly: (1) action of calcitriol on Ca intestinal absorption, (2) of PTH on calcitriol synthesis in the kidney, (3) of both on Ca reabsorption in the kidney, and (4) of Ca and calcitriol on PTH autoregulation. The link between the Ca/P homeostasis model and the bone remodelling model is PTH, which activates bone resorption.

The bone remodelling model considers tissue mineralization, which is controlled by Ca availability in bone marrow, a result from the homeostasis model. Phosphate homeostasis is crucial, as elevated serum P impairs renal calcitriol synthesis and contributes to the complex pathophysiology of CKD-associated bone disease.

CKD was simulated by a progressive reduction in glomerular filtration rate GFR from normal 6 L/h to 0.96 L/h [3]. pHPT was simulated by an abnormal production of PTH to reproduce clinical results [4]. The effect of diet on bone density was studied by varying Ca and P intake and vitamin D supplementation.

Results

Figs. 1 and 2 show the temporal evolution of bone density, normalized to its baseline value, in CKD and pHPT patients, respectively, under different dietary scenarios. Comparison with the nominal case (black lines) shows that increasing Ca intake has a negligible effect on bone density, whereas vitamin D supplementation attenuates bone loss, particularly in pHPT patients and, in CKD patients, when combined with reduced P intake. In contrast, increased P intake exacerbates bone loss in both conditions.

Conclusions

The proposed framework provides a quantitative tool to investigate the interplay between endocrine regulation, bone remodelling and dietary factors in metabolic bone diseases. In the future, it may support the design and optimization of pharmacological therapies by evaluating the potential benefits of adjunctive nutritional interventions.

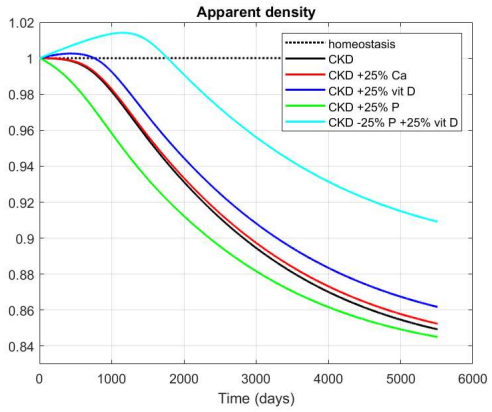


Figure 1: Effect of diet on CKD patients.

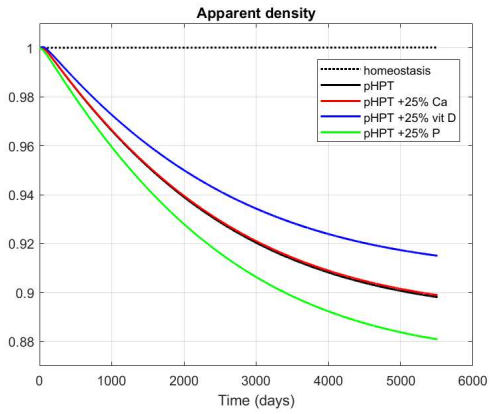


Figure 2: Effect of diet on pHPT patients.

Acknowledgements

The authors acknowledge Grant PID2024-161475OB-I00 funded by MICIU/AEI/ 10.13039/501100011033 and by "ERDF/EU".

References

[1] Martinez-Reina et al, Front Bioeng Biotechnol, 10: 886579, 2022.
[2] Peterson and Riggs, Bone, 46: 49-63, 2010.
[3] Rix et al, Kidney Int, 56: 1084-93, 1999.
[4] Rubin et al, J Clin Endocrinol Metab, 93: 3462, 2008.